

Chemical Reactions and Equations

- In a chemical reaction, at least one of the following will occur:
- Change in state
- Change in colour
- Evolution of a gas
- Change in temperature
- Formation of a precipitate

A **chemical equation** is the symbolic representation of a chemical reaction in the form of chemical formulae, signs, symbols, and directions. In which the reactant entities are given on the left-hand side and the product entities on the right-hand side.

- **Balanced chemical equation**

Reactants → Products

LHS
RHS

Total number of atoms on the LHS = Total number of atoms on the RHS

- How to balance an equation
- Write reactants and products
- Balance the maximum number of a particular atom on both sides
- Balance other atoms
- A complete balanced equation should look like



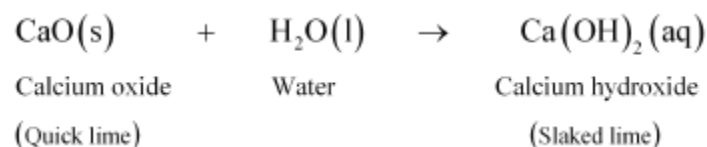
Types of reactions

- **Combination reaction**

- Two or more reactants combine to form one single product.

Examples: $\text{CaO s} + \text{H}_2\text{O l} \rightarrow \text{Ca(OH)}_2 \text{aq}$ Calcium oxide Water Calcium hydroxide (Quick lime) (Slaked lime) $\text{C s} + \text{O}_2 \text{g} \rightarrow \text{CO}_2 \text{g}$ Carbon Oxygen Carbon dioxide $2\text{H}_2 \text{g} + \text{O}_2 \text{g} \rightarrow 2\text{H}_2\text{O l}$ Hydrogen Oxygen Water

- **Exothermic reaction** – Heat gets released in the reaction. Most combination reactions are exothermic. For example,



- **Endothermic reaction** – Heat is absorbed in the reaction. Very few combination reactions are endothermic. For example,



- **Decomposition reaction**

- A single reactant breaks into several simple products.

Examples: 2FeSO_4 Ferrous sulphate $\xrightarrow{\Delta}$ Fe_2O_3 Ferric oxide + SO_2 + SO_3
 CaCO_3 Limestone $\xrightarrow{\Delta}$ CaO Calcium oxide + CO_2
 2AgCl Silver chloride $\xrightarrow{\Delta}$ 2Ag Silver + Cl_2

- All decomposition reactions are **endothermic [they absorb heat]**.

- **Displacement reactions:**

- In displacement reactions, a more reactive metal can displace a less reactive metal from their compounds in aqueous solutions. (However, a less reactive metal cannot displace a more reactive metal.)

Example: $\text{CuSO}_4 + \text{Zn} \rightarrow \text{ZnSO}_4 + \text{Cu}$
 Copper Sulphate (Blue) + Zinc (Red) → Zinc Sulphate (Colourless) + Copper
 $\text{Fe s} + \text{CuSO}_4\text{aq} \rightarrow \text{Cu s} + \text{FeSO}_4\text{aq}$
 Iron Copper sulphate → Copper Iron sulphate

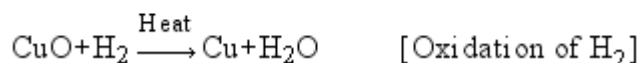
- **Double displacement reaction**

- Exchange of ions occurs between two compounds.

Example: $\text{Na}_2\text{SO}_4\text{aq} + \text{BaCl}_2\text{s} \rightarrow \text{BaSO}_4\text{aq} + 2\text{NaCl s}$
 Sodium sulphate + Barium chloride → Barium sulphate + Sodium chloride

- When the aqueous solution of two compounds react by exchanging their respective ions, such that one of the products formed is insoluble salt and appears in the form of a precipitate, then the reaction is said to be **precipitation reaction**.
- When an acid solution reacts with a base and the two exchange their respective ions, such that only salt and water are products, then the reaction is called **neutralisation reaction**.
- When two compounds react with each other and displace their ions, in such a manner that one of the product formed either decomposes into gaseous compounds or is formed in gaseous state, then the reaction is called **gas-forming reaction**.

- **Oxidation** → When a substance gains oxygen or loses hydrogen

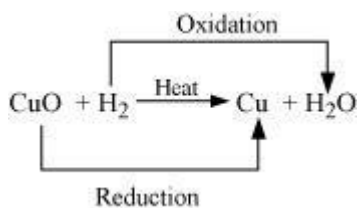


- **Oxidation in everyday life**

- **Corrosion** – When a metal is oxidised by the action of air and moisture [that's why metals are coated]
- **Rancidity** – When fats and oils are oxidised, their smell and taste change [that's why food is kept in air-tight containers]
- **Reduction** → When one substance loses oxygen or gains hydrogen



- **Redox** – Oxidation-reduction reaction



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